Basic Networking in Linux

IP address, subnet mask, gateway, DNS, LAN, MAN and WAN in simple words, with examples and Linux commands

Part	Topic	One-line idea
1	What is networking	Computers talking to each other
2	IP address	The address of a device
3	Subnet mask	Which devices are in my own network
4	Gateway	The exit door to other networks
5	DNS	The phone book that turns names into addresses
6	How they work together	The journey of one web request
7	LAN, MAN, WAN	Networks by size
8	Linux commands	Show it live to students
9	Troubleshooting ladder	Find which part is broken

1. What is Networking?

Networking means connecting computers and devices so they can **talk to each other and share data**. The network can be small (a LAN in your home) or huge (the Internet).

Everyday example	What is happening
Your laptop opens a website	Your laptop talks to a web server.
A Linux server uses a database	The server talks to a database server.
Two servers in a Kubernetes cluster	Servers send data to each other.

All of this works because of four basic parts: **IP address, subnet mask, gateway and DNS**.

2. IP Address

An **IP address** is a unique number given to every device on a network. It tells data **where to go**. Example: `192.168.1.10`

> **Analogy: Home address idea:** To send you a parcel, people need your home address. In the same way, to send data to a computer, you need its IP address.

Private IP and Public IP

	Private IP	Public IP
Used	Inside homes, offices and private networks	On the Internet

	Private IP	Public IP
Seen from the Internet?	No	Yes
Ranges	192.168.x.x 10.x.x.x 172.16.x.x to 172.31.x.x	Everything else
Examples	192.168.1.10, 10.0.0.5	8.8.8.8, 142.250.183.14

Trick: Check the first numbers. **192.168**, **10** or **172.16 to 172.31** = private. Anything else = public. (Careful: 172.32.x.x is public.)

3. Subnet Mask

A **subnet mask** splits an IP address into two parts: the **network part** (which network?) and the **host part** (which device inside it?).

```
IP address  : 192.168.1.25
Subnet mask : 255.255.255.0

Network part: 192.168.1     <-- same for every device in this network
Host part   : 25            <-- different for every device
```

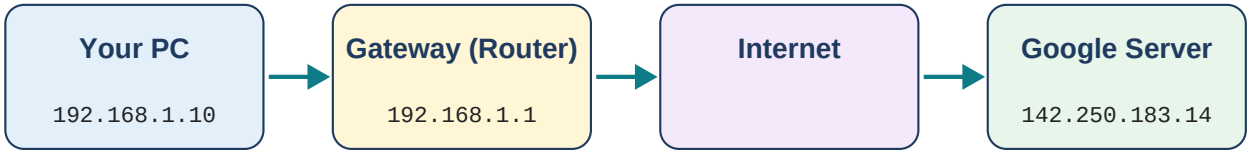
Where the mask has **255**, that part of the IP is the **network**. Where it has **0**, that part is the **host**. All devices that start with `192.168.1.` are in the **same network**. The mask 255.255.255.0 is also written **/24**.

Device	Same network as 192.168.1.25?	Why
192.168.1.50	Yes	Network part 192.168.1 is the same
192.168.1.200	Yes	Network part 192.168.1 is the same
192.168.2.50	No	Network part is 192.168.2, so it is different

Analogy: Street and house: the **network part is the street name** and the **host part is the house number**. Houses on the same street are neighbours and can talk directly. A house on another street needs a road out of the street (the gateway).

4. Gateway (Default Gateway)

A **gateway** is the **router** that connects your local network to other networks, such as the Internet. It is the **main exit gate** of your office or apartment.



Google is **outside** your network, so your PC sends the data to the gateway first, and the gateway forwards it. **Without a gateway, your computer can talk only to devices in the same local network.**

COPY AND PASTE: Find your gateway

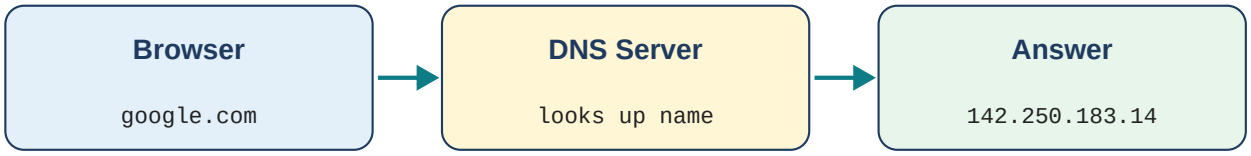
```
ip route
```

Output (what students will see)

```
default via 192.168.1.1 dev eth0     <-- 192.168.1.1 is the default gateway
192.168.1.0/24 dev eth0 scope link    <-- the local network
```

5. DNS (Domain Name System)

People remember **names** better than **numbers**. We type `google.com`, not `142.250.183.14`. **DNS** is like a phone book: it converts a domain name into its IP address.



COPY AND PASTE: Ask DNS for an IP address

```
nslookup google.com
dig google.com
dig +short google.com
```

Command	What it does (in simple words)
`nslookup google.com`	Asks DNS for the IP address of google.com.
`dig google.com`	Same, with full details (record type A = IPv4 address).
`dig +short google.com`	Prints only the IP address.

Output (what students will see)

```
Name:    google.com
Address: 142.250.183.14               <-- nslookup answer

google.com.   300   IN   A   142.250.183.14   <-- dig: A record
142.250.183.14                        <-- dig +short
```

Tip: If **dig** is not found, install it: `sudo apt install -y dnsutils` (Red Hat: `sudo dnf install -y bind-utils`).

6. How They All Work Together

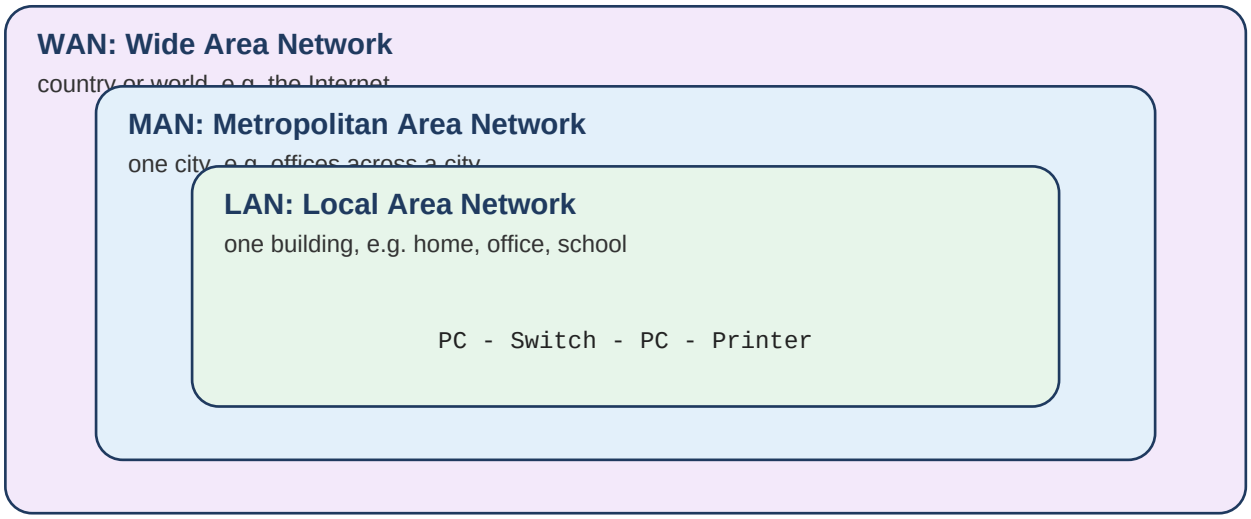
What happens when you open **https://google.com**:

1. **Browser asks DNS**
   'What is the IP address of google.com?'
2. **DNS replies**
   142.250.183.14
3. **Computer checks**
   'Is this IP in my local network?' (uses the subnet mask) Answer: No
4. **Packet goes to the Default Gateway**
   The router at 192.168.1.1
5. **Gateway forwards it through the Internet**
   to Google's server
6. **Google sends the web page back**
   the same way, and the browser shows it

Component	Purpose
IP address	Identifies a device on the network.
Subnet mask	Separates the network part from the host part.
Gateway	Sends traffic from the local network to other networks.
DNS	Converts domain names into IP addresses.

7. LAN, MAN and WAN

These are types of networks, named by the **area they cover**.



Feature	LAN	MAN	WAN
Full form	Local Area Network	Metropolitan Area Network	Wide Area Network

Feature	LAN	MAN	WAN
Coverage	Home, office, school, building	One city	Country or whole world
Speed	High	Medium to high	Lower than LAN
Cost	Low	Medium	High
Owned by	One organization	Organization or ISP	ISPs and telecom providers
Example	Office Wi-Fi, college lab	Company offices across a city	The Internet

Network	Easy example
LAN	PCs, a printer and a file server connected to one switch in an office. They share files, printers and the Internet.
MAN	A company has Office A, B and C in the same city, linked by high-speed fibre.
WAN	A company has offices in India, USA and UK, connected over the Internet. The Internet is the world's largest WAN.

Trick: LAN = one building. **MAN** = one city. **WAN** = many cities or countries.

8. Linux Commands

Command	What it does
`ip addr`	Show IP addresses of network interfaces
`hostname -I`	Show the system's IP address
`ip route`	Show the routing table and default gateway
`ping google.com`	Test network connectivity (use -c 3 to send only 3)
`curl https://google.com`	Test HTTP/HTTPS connectivity
`dig google.com`	Query DNS records
`nslookup google.com`	Resolve a domain name to an IP address

COPY AND PASTE: Network commands (copy and paste)

```
ip addr
hostname -I
ip route
ping -c 3 google.com
curl -I https://google.com
dig +short google.com
```

Command	What it does (in simple words)
`ip addr`	Look for the line starting with inet: that is your IP and mask.

Command	What it does (in simple words)
`hostname -I`	Only the IP address.
`ip route`	The line starting with default via shows the gateway.
`ping -c 3 google.com`	Sends 3 test packets. Replies mean the Internet and DNS work.
`curl -I https://google.com`	Fetches only the web headers: proves web access works.
`dig +short google.com`	Shows the IP address DNS gives for the name.

Output (what students will see)

```
    inet 192.168.1.10/24 brd 192.168.1.255 scope global eth0   <-- my IP / mask
192.168.1.10                                  <-- hostname -I
default via 192.168.1.1 dev eth0              <-- gateway
64 bytes from 142.250.183.14: icmp_seq=1 ttl=117 time=18.2 ms  <-- ping reply
HTTP/2 301                                    <-- curl: site answered
142.250.183.14                                <-- dig +short
```

9. Troubleshooting Ladder: Which Part is Broken?

Test step by step, from your own computer outwards. The **first test that fails** shows where the problem is.

COPY AND PASTE: Step-by-step network test

```
ping -c 2 127.0.0.1
ping -c 2 $(ip route | awk '/default/ {print $3}')
ping -c 2 8.8.8.8
ping -c 2 google.com
```

Command	What it does (in simple words)
`ping -c 2 127.0.0.1`	Pings your own computer (loopback). Tests the network software.
`ping -c 2 $(ip route | awk '/default/ {print $3}')`	Pings your gateway. The part inside $( ) finds the gateway IP from ip route.
`ping -c 2 8.8.8.8`	Pings a public IP. Tests the Internet without DNS.
`ping -c 2 google.com`	Pings by name. Tests DNS as well.

If this fails	The problem is
ping 127.0.0.1	Network software on your computer.
ping gateway	Your local network, cable, Wi-Fi or router.
ping 8.8.8.8 (gateway worked)	Router or Internet connection (ISP).
ping google.com (8.8.8.8 worked)	**DNS**: names are not resolving.

10. Key Takeaways and Quick Quiz

- Every device on a network has an **IP address**.
- The **subnet mask** shows the network part and the host part of an IP.
- The **gateway** lets devices reach other networks and the Internet.
- **DNS** translates domain names into IP addresses.
- Together they make communication over the Internet possible.
- **LAN** is one building, **MAN** is one city, **WAN** is the world.

Quick quiz

Question	Answer
What does an IP address do?	Identifies a device and tells data where to go.
Is 172.20.5.5 private or public?	Private (172.16 to 172.31).
With mask 255.255.255.0, are 192.168.1.5 and 192.168.1.90 in the same network?	Yes: both have the network part 192.168.1.
What does the gateway do?	Forwards traffic from your network to other networks.
Which command shows the gateway?	ip route
What does DNS do?	Converts a domain name into an IP address.
ping 8.8.8.8 works but ping google.com fails. Why?	DNS problem.
Which network type is the Internet?	WAN (the world's largest).